

424 D LS

ISTRUZIONI PER L'USO - NORME DI INSTALLAZIONE
INSTRUCTIONS FOR USE - DIRECTIONS FOR INSTALLATION
INSTRUCTIONS - REGLES D'INSTALLATION
INSTRUCCIONES PARA EL USO - NORMAS PARA LA INSTALACION
GEBRAUCHSANLEITUNG - ANWEISUNGEN ZUR INSTALLATION
GEbruikSAANWIJZINGEN – INSTALLATIEVOORSCHRIFTEN

FAAC



AVVERTENZE PER L'INSTALLATORE

OBBLIGHI GENERALI PER LA SICUREZZA

1) ATTENZIONE! È importante per la sicurezza delle persone seguire attentamente tutta l'istruzione. Una errata installazione o un errato uso del prodotto può portare a gravi danni alle persone.

- 2) Leggere attentamente le istruzioni prima di iniziare l'installazione del prodotto.
- 3) I materiali dell'imballaggio (plastica, polistirolo, ecc.) non devono essere lasciati alla portata dei bambini in quanto potenziali fonti di pericolo.
- 4) Conservare le istruzioni per riferimenti futuri.
- 5) Questo prodotto è stato progettato e costruito esclusivamente per l'utilizzo indicato in questa documentazione. Qualsiasi altro utilizzo non espressamente indicato potrebbe pregiudicare l'integrità del prodotto e/o rappresentare fonte di pericolo.
- 6) FAAC declina qualsiasi responsabilità derivata dall'uso improprio o diverso da quello per cui l'automatismo è destinato.
- 7) Non installare l'apparecchio in atmosfera esplosiva: la presenza di gas o fumi infiammabili costituisce un grave pericolo per la sicurezza.
- 8) Gli elementi costitutivi meccanici devono essere in accordo con quanto stabilito dalle Norme EN 12604 e EN 12605.

Per i Paesi extra-CEE, oltre ai riferimenti normativi nazionali, per ottenere un livello di sicurezza adeguato, devono essere seguite le Norme sopra riportate.

9) FAAC non è responsabile dell'inosservanza della Buona Tecnica nella costruzione delle chiusure da motorizzare, nonché delle deformazioni che dovessero intervenire nell'utilizzo.

10) L'installazione deve essere effettuata nell'osservanza delle Norme EN 12453 e EN 12445. Il livello di sicurezza dell'automazione deve essere C+D.

11) Prima di effettuare qualsiasi intervento sull'impianto, togliere l'alimentazione elettrica e scollegare le batterie.

12) Prevedere sulla rete di alimentazione dell'automazione un interruttore onnipolare con distanza d'apertura del contatto uguale o superiore a 3 mm. È consigliabile l'uso di un magnetotermico da 6A con interruzione onnipolare.

13) Verificare che a monte dell'impianto vi sia un interruttore differenziale con soglia da 0,03 A.

14) Verificare che l'impianto di terra sia realizzato a regola d'arte e collegarvi le parti metalliche della chiusura.

15) L'automazione dispone di una sicurezza intrinseca antischiacciamento costituita da un controllo di coppia. E' comunque necessario verificare la soglia di intervento secondo quanto previsto dalle Norme indicate al punto 10.

16) I dispositivi di sicurezza (norma EN 12978) permettono di proteggere eventuali aree di pericolo da rischi meccanici di movimento, come ad Es. schiacciamento, coinvolgimento, cesoiamento.

17) Per ogni impianto è consigliato l'utilizzo di almeno una segnalazione luminosa nonché di un cartello di segnalazione fissato adeguatamente sulla struttura dell'infisso, oltre ai dispositivi citati al punto "16".

18) FAAC declina ogni responsabilità ai fini della sicurezza e del buon funzionamento dell'automazione, in caso vengano utilizzati componenti dell'impianto non di produzione FAAC.

19) Per la manutenzione utilizzare esclusivamente parti originali FAAC.

20) Non eseguire alcuna modifica sui componenti facenti parte del sistema d'automazione.

21) L'installatore deve fornire tutte le informazioni relative al funzionamento manuale del sistema in caso di emergenza e consegnare all'Utente utilizzatore dell'impianto il libretto d'avvertenze allegato al prodotto.

22) Non permettere ai bambini o persone di sostare nelle vicinanze del prodotto durante il funzionamento.

23) Tenere fuori dalla portata dei bambini (e diocorandoli o qualsiasi altro datore di impulso, per evitare che l'automazione possa essere azionata involontariamente).

24) Il transito tra le ante deve avvenire solo a cancello completamente aperto.

25) L'Utente utilizzatore deve astenersi da qualsiasi tentativo di riparazione o d'intervento diretto e rivolgersi solo a personale qualificato.

26) Tutto quello che non è previsto espressamente in queste istruzioni non è permesso

IMPORTANT NOTICE FOR THE INSTALLER

GENERAL SAFETY REGULATIONS

1) ATTENTION! To ensure the safety of people, it is important that you read all the following instructions. Incorrect installation or incorrect use of the product could cause serious harm to people.

- 2) Carefully read the instructions before beginning to install the product.
- 3) Do not leave packing materials (plastic, polystyrene, etc.) within reach of children as such materials are potential sources of danger.
- 4) Store these instructions for future reference.
- 5) This product was designed and built strictly for the use indicated in this documentation. Any other use, not expressly indicated here, could compromise the good condition/operation of the product and/or be a source of danger.
- 6) FAAC declines all liability caused by improper use or use other than that for which the automated system was intended.
- 7) Do not install the equipment in an explosive atmosphere: the presence of inflammable gas or fumes is a serious danger to safety.
- 8) The mechanical parts must conform to the provisions of Standards EN 12604 and EN 12605.

For non-EU countries, to obtain an adequate level of safety, the Standards mentioned above must be observed, in addition to national legal regulations.

9) FAAC is not responsible for failure to observe Good Technique in the construction of the closing elements to be motorised, or for any deformation that may occur during use.

10) The installation must conform to Standards EN 12453 and EN 12445. The safety level of the automated system must be C+D.

11) Before effecting any job on the system, cut out electrical power and disconnect the batteries.

12) The mains power supply of the automated system must be fitted with an all-pole switch with contact opening distance of 3mm or greater. Use of a 6A thermal breaker with all-pole circuit break is recommended.

13) Make sure that a differential switch with threshold of 0.03 A is fitted upstream of the system.

14) Make sure that the earthing system is perfectly constructed, and connect metal parts of the means of the closure to it.

15) The automated system is supplied with an intrinsic anti-crushing safety device

consisting of a torque control. Nevertheless, its tripping threshold must be checked as specified in the Standards indicated at point 10.

- 16) The safety devices (EN 12978 standard) protect any danger areas against mechanical movement Risks, such as crushing, dragging, and shearing.
- 17) Use of at least one indicator-light is recommended for every system, as well as a warning sign adequately secured to the frame structure, in addition to the devices mentioned at point "16".
- 18) FAAC declines all liability as concerns safety and efficient operation of the automated system, if system components not produced by FAAC are used.
- 19) For maintenance, strictly use original parts by FAAC.
- 20) Do not in any way modify the components of the automated system.
- 21) The installer shall supply all information concerning manual operation of the system in case of an emergency, and shall hand over to the user the warnings handbook supplied with the product.
- 22) Do not allow children or adults to stay near the product while it is operating.
- 23) Keep remote controls or other pulse generators away from children, to prevent the automated system from being activated involuntarily.
- 24) Transit through the leaves is allowed only when the gate is fully open.
- 25) The user must not attempt any kind of repair or direct action whatever and contact qualified personnel only.
- 26) Anything not expressly specified in these instructions is not permitted.

CONSIGNES POUR L'INSTALLATEUR

RÈGLES DE SÉCURITÉ

1) ATTENTION! Il est important, pour la sécurité des personnes, de suivre à la lettre toutes les instructions. Une installation erronée ou un usage erroné du produit peut entraîner de graves conséquences pour les personnes.

- 2) Lire attentivement les instructions avant d'installer le produit.
- 3) Les matériaux d'emballage (matière plastique, polystyrène, etc.) ne doivent pas être laissés à la portée des enfants car ils constituent des sources potentielles de danger.
- 4) Conserver les instructions pour les références futures.
- 5) Ce produit a été conçu et construit exclusivement pour l'usage indiqué dans cette documentation. Toute autre utilisation non expressément indiquée pourrait compromettre l'intégrité du produit et/ou représenter une source de danger.
- 6) FAAC décline toute responsabilité qui dériverait d'usage impropre ou différent de celui auquel l'automatisme est destiné.
- 7) Ne pas installer l'appareil dans une atmosphère explosive: la présence de gaz ou de fumées inflammables constitue un grave danger pour la sécurité.
- 8) Les composants mécaniques doivent répondre aux prescriptions des Normes EN 12604 et EN 12605.

Pour les Pays extra-CEE, l'obtention d'un niveau de sécurité approprié exige non seulement le respect des normes nationales, mais également le respect des Normes surmentionnées.

9) FAAC n'est pas responsable du non-respect de la Bonne Technique dans la construction des fermetures à motoriser, ni des déformations qui pourraient intervenir lors de l'utilisation.

10) L'installation doit être effectuée conformément aux Normes EN 12453 et EN 12445. Le niveau de sécurité de l'automatisme doit être C+D.

11) Couper l'alimentation électrique et déconnecter la batterie avant toute intervention sur l'installation.

12) Prévoir, sur le secteur d'alimentation de l'automatisme, un interrupteur onnipolaire avec une distance d'ouverture des contacts égale ou supérieure à 3 mm. On recommande d'utiliser un magnétothermique de 6A avec interruption onnipolaire.

13) Vérifier qu'il y ait, en amont de l'installation, un interrupteur différentiel avec un seuil de 0,03 A.

14) Vérifier que la mise à terre est réalisée selon les règles de l'art et y connecter les pièces métalliques de la fermeture.

15) L'automatisme dispose d'une sécurité intrinsèque anti-écrasement, formée d'un contrôle du couple. Il est toutefois nécessaire d'en vérifier le seuil d'intervention suivant les prescriptions des Normes indiquées au point 10.

16) Les dispositifs de sécurité (norme EN 12978) permettent de protéger des zones éventuellement dangereuses contre les Risques mécaniques du mouvement, comme l'écrasement, l'acheminement, le cisaillement.

17) On recommande que toute installation soit dotée au moins d'une signalisation lumineuse, d'un panneau de signalisation fixe, de manière appropriée, sur la structure de la fermeture, ainsi que des dispositifs cités au point "16".

18) FAAC décline toute responsabilité quant à la sécurité et au bon fonctionnement de l'automatisme si les composants utilisés dans l'installation n'appartiennent pas à la production FAAC.

19) Utiliser exclusivement, pour l'entretien, des pièces FAAC originales.

20) Ne jamais modifier les composants faisant partie du système d'automatisme.

21) L'installateur doit fournir toutes les informations relatives au fonctionnement manuel du système en cas d'urgence et remettre à l'Usager qui utilise l'installation les "instructions pour l'Usager" fournies avec le produit.

22) Interdire aux enfants ou aux tiers de stationner près du produit durant le fonctionnement.

23) Eloigner de la portée des enfants les radiocommandes ou tout autre générateur d'impulsions, pour éviter tout actionnement involontaire de l'automatisme.

24) Le transit entre les vantaux ne doit avoir lieu que lorsque le portail est complètement ouvert.

25) L'Usager qui utilise l'installation doit éviter toute tentative de réparation ou d'intervention directe et s'adresser uniquement à un personnel qualifié.

26) Tout ce qui n'est pas prévu expressément dans ces instructions est interdit.

ADVERTENCIAS PARA EL INSTALADOR

REGLAS GENERALES PARA LA SEGURIDAD

1) ATENCION! Es sumamente importante para la seguridad de las personas seguir atentamente las presentes instrucciones. Una instalación incorrecta o un uso impropio del producto puede causar graves daños a las personas.

- 2) Leer detenidamente las instrucciones antes de instalar el producto.
- 3) Los materiales del embalaje (plástico, poliestireno, etc.) no deben dejarse al alcance de los niños, ya que constituyen fuentes potenciales de peligro.
- 4) Guardar las instrucciones para futuras consultas.
- 5) Este producto ha sido proyectado y fabricado exclusivamente para la utilización indicada en el presente manual. Cualquier uso diverso del previsto podría perjudicar el funcionamiento del producto y/o representar fuente de peligro.
- 6) FAAC declina cualquier responsabilidad derivada de un uso impropio o diverso

- del previsto.
- 7) No instalen el aparato en atmósfera explosiva; la presencia de gas o humos inflamables constituye un grave peligro para la seguridad.
 - 8) Los elementos constructivos mecánicos deben estar de acuerdo con lo establecido en las Normas EN 12604 y EN 12605.
 - Para los países no pertenecientes a la CEE, además de las referencias normativas nacionales, para obtener un nivel de seguridad adecuado, deben seguirse las Normas arriba indicadas.
 - 9) FAAC no es responsable del incumplimiento de las buenas técnicas de fabricación de los clientes que se han de motorizar, así como de las deformaciones que pudieran intervenir en la utilización.
 - 10) La instalación debe ser realizada de conformidad con las Normas EN 12453 y EN 12445. El nivel de seguridad de la automatización debe ser C+D.
 - 11) Quitar la alimentación eléctrica y desconectar las baterías antes de efectuar cualquier intervención en la instalación.
 - 12) Colocar en la red de alimentación de la automatización un interruptor omnipolar con distancia de apertura de los contactos igual o superior a 3 mm. Se aconseja usar un magnetotérmico de 6A con interrupción omipolar.
 - 13) Comprueben que la instalación disponga línea arieta de un interruptor diferencial con umbral de 0,03 A.
 - 14) Verifiquen que la instalación de tierra esté correctamente realizada y conecten las partes metálicas del cierre.
 - 15) La automatización dispone de un dispositivo de seguridad antiaplasmiento constituido por un control de par. No obstante, es necesario comprobar el umbral de intervención según lo previsto en las Normas indicadas en el punto 10.
 - 16) Los dispositivos de seguridad (norma EN 12978) permiten proteger posibles áreas de peligro de Riesgos mecánicos de movimiento, como por ej. aplastamiento, arrastre, corte.
 - 17) Para cada equipo se aconseja usar por lo menos una señalización luminosa así como un cartel de señalización adecuadamente fijado a la estructura del bastidor, además de los dispositivos indicados en el "16".
 - 18) FAAC declina toda responsabilidad relativa a la seguridad y al buen funcionamiento de la automatización si se utilizan componentes de la instalación que no sean de producción FAAC.
 - 19) Para el mantenimiento utilicen exclusivamente piezas originales FAAC
 - 20) No efectúen ninguna modificación en los componentes que forman parte del sistema de automatización.
 - 21) El instalador debe proporcionar todas las informaciones relativas al funcionamiento del sistema en caso de emergencia y entregar al usuario del equipo el manual de advertencias que se adjunta al producto.
 - 22) No permitan que niños o personas se defiengan en proximidad del producto durante su funcionamiento.
 - 23) Mantengan lejos del alcance los niños los telemandos o cualquier otro emisor de impulso, para evitar que la automatización pueda ser accionada involuntariamente.
 - 24) Sólo puede transferirse entre las hojas si la cancela está completamente abierta.
 - 25) El usuario no debe por ningún motivo intentar reparar o modificar el producto, debe siempre dirigirse a personal cualificado.
 - 26) Todo lo que no esté previsto expresamente en las presentes instrucciones debe entenderse como no permitido

HINWEISE FÜR DEN INSTALLATIONSTECHNIKER ALLGEMEINE SICHERHEITSVORSCHRIFTEN

- 1) **ACHTUNG!** Um die Sicherheit von Personen zu gewährleisten, sollte die Anleitung aufmerksam befolgt werden. Eine falsche Installation oder ein fehlerhafter Betrieb des Produktes können zu schwerwiegenden Personenschäden führen.
- 2) Bevor mit der Installation des Produktes begonnen wird, sollten die Anleitungen aufmerksam gelesen werden.
- 3) Das Verpackungsmaterial (Kunststoff, Styropor, usw.) sollte nicht in Reichweite von Kindern aufbewahrt werden, da es eine potentielle Gefahrenquelle darstellt.
- 4) Die Anleitung sollte aufbewahrt werden, um auch in Zukunft Bezug auf sie nehmen zu können.
- 5) Dieses Produkt wurde ausschließlich für den in diesen Unterlagen angegebenen Gebrauch entwickelt und hergestellt. Jeder andere Gebrauch, der nicht ausdrücklich angegeben ist, könnte die Unversehrtheit des Produktes beeinträchtigen und/oder eine Gefahrenquelle darstellen.
- 6) Die Firma FAAC lehnt jede Haftung für Schäden, die durch unsachgemäßen oder nicht bestimmungsgemäßen Gebrauch der Automatik verursacht werden, ab.
- 7) Das Gerät sollte nicht in explosionsgefährdeten Umgebungen installiert werden: das Vorhandensein von entflammaren Gasen oder Rauch stellt ein schwerwiegendes Sicherheitsrisiko dar.
- 8) Die mechanischen Bauelemente müssen den Anforderungen der Normen EN 12604 und EN 12605 entsprechen.
- Für Länder, die nicht der Europäischen Union angehören, sind für die Gewährleistung eines entsprechenden Sicherheitsniveaus neben den nationalen gesetzlichen Bezugsvorschriften die oben aufgeführten Normen zu beachten.
- Die Firma FAAC übernimmt keine Haftung im Falle von nicht fachgerechten Ausführungen bei der Herstellung der anzubauenden Schließvorrichtungen sowie bei Deformationen, die eventuell beim Betrieb entstehen.
- 10) Die Installation muß unter Beachtung der Normen EN 12453 und EN 12445 erfolgen. Die Sicherheitsstufe der Automatik sollte C+D sein.
- 11) Vor der Ausführung jeglicher Eingriffe auf der Anlage sind die elektrische Versorgung und die Batterie abzunehmen.
- 12) Auf dem Versorgungsnetz der Automatik ist ein unipolarer Schalter mit Öffnungsabstand der Kontakte von über oder gleich 3 mm einzubauen. Darüber hinaus wird der Einsatz eines Magnetschutzschalters mit 6A mit unipolarer Abschaltung empfohlen.
- 13) Es sollte überprüft werden, ob vor der Anlage ein Differentialschalter mit einer Auslöseschwelle von 0,03 A zwischengeschaltet ist.
- 14) Es sollte überprüft werden, ob die Erdungsanlage fachgerecht ausgeführt wurde. Die Metallteile der Schließung sollten an diese Anlage angeschlossen werden.
- 15) Die Automation verfügt über eine eingebaute Sicherheitsvorrichtung für den Guteschutz, die aus einer Drehmomentkontrolle besteht. Es ist in jedem Falle erforderlich, diesen Eingriffsschwellen gemäß der Vorgaben der unter Punkt 10 angegebenen Vorschriften zu überprüfen.
- 16) Die Sicherheitsvorrichtungen (Norm EN 12978) ermöglichen den Schutz eventueller Gefahrenbereiche vor mechanischen Bewegungsschritten, wie zum Beispiel Quetschungen, Mitschleifen oder Schnittverletzungen.
- 17) Für jede Anlage wird der Einsatz von mindestens einem Leuchtsignal empfohlen

- sowie eines Hinweisschildes, das über eine entsprechende Befestigung mit dem Aufbau des Tor verbunden wird. Darüber hinaus sind die unter Punkt "16" erwähnten Vorrichtungen einzusetzen.
- 18) Die Firma FAAC lehnt jede Haftung hinsichtlich der Sicherheit und des störungsfreien Betriebs der Automatik ab, soweit Komponenten auf der Anlage eingesetzt werden, die nicht im Hause FAAC hergestellt wurden.
 - 19) Bei der Instandhaltung sollten ausschließlich Originalteile der Firma FAAC verwendet werden.
 - 20) Auf den Komponenten, die Teil des Automationssystems sind, sollten keine Veränderungen vorgenommen werden.
 - 21) Der Installateur sollte alle Informationen hinsichtlich des manuellen Betriebs des Systems in Notfällen liefern und dem Betreiber der Anlage das Anleitungsbuch, das dem Produkt beigelegt ist, übergeben.
 - 22) Weder Kinder noch Erwachsene sollten sich während des Betriebs in der unmittelbaren Nähe der Automatik aufhalten.
 - 23) Die Funktionseuerungen und alle anderen Impulsgeber sollten außerhalb der Reichweite von Kindern aufbewahrt werden, um ein versehentliches Aktivieren der Automation zu vermeiden.
 - 24) Der Durchgang oder die Durchfahrt zwischen den Flügeln darf lediglich bei vollständig geöffnetem Tor erfolgen.
 - 25) Der Betreiber sollte keinerlei Reparaturen oder direkte Eingriffe auf der Automation ausführen, sondern sich hierfür ausschließlich an qualifiziertes Fachpersonal wenden.
 - 26) Alle Vorgehensweisen, die nicht ausdrücklich in der vorliegenden Anleitung vorgesehen sind, sind nicht zulässig

WAARSCHUWINGEN VOOR DE INSTALLATEUR ALGEMENE VEILIGHEIDSVORSCHRIFTEN

- 1) **LET OP!** Het is belangrijk voor de veiligheid dat deze hele instructie zorgvuldig wordt opgevolgd. Een onjuiste installatie of foutief gebruik van het product kunnen ernstig persoonlijk letsel veroorzaken.
- 2) Lees de instructies aandachtig door alvorens te beginnen met de installatie van het product.
- 3) De verpakkingsmaterialien (plastic, polystyreen, enz.) mogen niet binnen het bereik van kinderen worden gelaten, want zij vormen een mogelijke bron van gevaar.
- 4) Bewaar de instructies voor raadpleging in de toekomst.
- 5) Dit product is uitsluitend ontworpen en gebouwd voor het doel dat in deze documentatie wordt aangegeven. Elk ander gebruik, dat niet uitdrukkelijk wordt vermeld, zou het product kunnen beschadigen en/of een bron van gevaar kunnen vormen.
- 6) FAAC aanvaardt geen enkele aansprakelijkheid voor schade die voortvloeit uit oneigenlijk gebruik of ander gebruik dan waarvoor het automatische systeem is bedoeld.
- 7) Installeer het apparaat niet in een explosiegevaarlijke omgeving; de aanwezigheid van ontvlambare gassen of dampen vormt een ernstig gevaar voor de veiligheid.
- 8) De mechanische bouwlementen moeten in overeenstemming zijn met de bepalingen van de normen EN 12604 en EN 12605.
- Voor niet-EEG landen moeten, om een goed veiligheidsniveau te bereiken, behalve de nationale voorschriften ook de bovenstaande normen in acht worden genomen.
- 9) FAAC is niet aansprakelijk als de regels der goede techniek niet in acht genomen zijn bij de bouw van het sluitwerk dat gemotoriseerd moet worden, noch voor vervormingen die zouden kunnen ontstaan bij het gebruik.
- 10) De installatie dient te geschieden in overeenstemming met de normen EN 12453 en EN 12445. Het veiligheidsniveau van het automatische systeem moet C+D zijn.
- 11) Alvorens ingrepen te gaan verrichten op de installatie moet de elektrische voeding van het systeem worden weggenomen en de batterijen worden afgekoppeld.
- 12) Zorg op het voedingsnet van het automatische systeem voor een meerpolige schakelaar met een opening tussen de contacten van 3 mm of meer. Het wordt geadviseerd een magnetothermische schakelaar van 6A te gebruiken met meerpolige onderbreking.
- 13) Controleer of er bovenstrooms van de installatie een differentieel-schakelaar is geïnstalleerd met een limiet van 0,03 A.
- 14) Controleer of de aardinginstallatie voldoende is aangelegd en sluit er de metalen delen van het sluitsysteem op aan.
- 15) Het automatische systeem beschikt over een intrinsieke beveiliging tegen inklemming, bestaande uit een controle van het koppel. De inschakelmoment hiervan dient echter te worden gecontroleerd volgens de bepalingen van de normen die worden vermeld onder punt 10.
- 16) De veiligheidsvoorzieningen (norm EN 12978) maken het mogelijk eventuele gevaarlijke gebieden te beschermen tegen Mechanische gevaren door beweging, zoals bijvoorbeeld inklemming, meesleuren of amputatie.
- 17) Het wordt voor elke installatie geadviseerd minstens één lichtsignaal te gebruiken alsook een waarschuwingbord dat goed op de constructie van het hang- en sluitwerk dient te worden bevestigd, afgezien nog van de voorzieningen die genoemd zijn onder punt "16".
- 18) FAAC aanvaardt geen enkele aansprakelijkheid voor wat betreft de veiligheid en de goede werking van het automatische systeem, als er in de installatie gebruik gemaakt wordt van componenten die niet door FAAC zijn geproduceerd.
- 19) Gebruik voor het onderhoud uitsluitend originele FAAC-onderdelen.
- 20) Vericht geen wijzigingen op componenten die deel uitmaken van het automatische systeem.
- 21) De installateur dient alle informatie te verstrekken over de handbediening van het systeem in noodgevallen, en moet de gebruiker van de installatie bij het product geleverde boekje met aanwijzingen overhandigen.
- 22) Sla het niet toe dat kinderen of volwassenen zich ophouden in de buurt van het product terwijl dit in werking is.
- 23) Houd radio-afstandsbedieningen of alle andere impulsgevers buiten het bereik van kinderen, om te voorkomen dat het automatische systeem onopzettelijk kan worden aangedreven.
- 24) Ga alleen tussen de vleugels door als het hek helemaal geopend is.
- 25) De gebruiker mag geen pogingen tot reparatie doen of direct ingrepen plegen, en dient zich uitsluitend te wenden tot gekwalificeerd personeel.
- 26) Alles wat niet uitdrukkelijk in deze instructies wordt aangegeven, is niet toegestaan

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CE DECLARATION OF CONFORMITY

Manufacturer: FAAC S.p.A.

Address: Via Benini, 1 - 40069 - Zola Predosa - Bologna - ITALY

Declares that: Control unit mod. **424 D LS**

- conforms to the essential safety requirements of the following EEC directives:
73/23/EEC and subsequent amendment 93/68/EEC.
89/336/EEC and subsequent amendment 92/31/EEC and 93/68/EEC

Additional information:

This product underwent a test in a typical uniform configuration (all products manufactured by FAAC S.p.A.).

Bologna, 10-01-2007

Mandating Director
A. Bassi

Notes on reading the instruction

Read this installation manual to the full before you begin installing the product.

The symbol  indicates notes that are important for the safety of persons and for the good condition of the automated system.

The symbol  draws your attention to the notes on the characteristics and operation of the product.

24 Vdc ELECTRONIC CONTROL UNIT FOR SWING GATES

USE INSTRUCTIONS - INSTALLATION INSTRUCTIONS

1. GENERAL CHARACTERISTICS

This 24 Vdc control unit for swing gates offers high performance and a wide range of adjustments: opening and closing decelerations, possibility of managing one or two motors, management of opening and closing limit-switches, and the possibility of managing two **GATECODERS**.

A sophisticated electronic control constantly monitors the power circuit and disables the control unit in the event of malfunctions that could impair efficiency of the electronic clutch.

The parameter settings and the operating logics are set and shown on a handy display, which indicates gate status during normal operation. Operating times are adjusted by self-learning during programming.

The water-tight enclosure is designed to house the control unit, the toroidal transformer and any buffer batteries (optional) having the characteristics and dimensions indicated in the table below.

2. TECHNICAL SPECIFICATIONS

Supply voltage of transformer	230/115 V~ (+6 -10%) - 50/60 Hz.
Supply voltage of control unit	24 V~ (+6 -10%) - 50/60 Hz.
Absorbed power	3 W
Motor max load	70 W x 2
Accessories max load	24Vdc 500mA
Flashing lamp/Courtesy light max. load	24Vdc 15W max.
Operating ambient temperature	-20°C +50°C
Protection fuses	4
Function logics	Automatic / Stepped Automatic / Semiautomatic / Stepped Semiautomatic / Condo type
Opening / closing time	Through self-learning during programming
Pause time	Through self-learning during programming
Thrust force	Four levels adjustable on display
Decelerations	Opening and closing Power supply 24V~ / Battery supply / Encoder
Terminal board inputs	Total opening / Pedestrian opening / Opening-closing safety devices / Stop / Opening-closing limit-switch
Radio connector	Rapid 5-pin connector
Terminal board outputs	24Vdc power supply to accessories / 24 Vdc Motors / 24 Vdc Courtesy light-Flashing lamp / 12 Vdc/~ Electric lock
Board dimensions	165 x 130 mm.
Characteristics of 230V~ toroidal transformer	prim. 230V~ / sec. 22V~ / 120VA
Characteristics of 115V~ toroidal transformer	prim. 115V~ / sec. 20V~ / 120VA
Characteristics of optional batteries	12V - 4 Ah / dimens. 90 x 70 x 108 mm.
Characteristics of outdoor enclosure	306 x 225 x 130 mm. - IP55



Different output values can be obtained on the 24V~ voltage depending on the mains voltage value. Before start-up, always check the transformer output voltage. This must not exceed 26V~ for both 230V~ and 115V~ power supply. Voltage must be measured load free, i.e. with the transformer powered and disconnected from the board.

3. PRELIMINARY SETTING-UP



To ensure people's safety, all warnings and instructions in this booklet must be carefully observed. Incorrect installation or incorrect use of the product could cause serious harm to people.

Make sure there is an adequate differential switch upstream of the system as specified by current laws, and install an all-pole thermal breaker on the power supply mains.

To lay electric cables, use adequate rigid and/or flexible pipes.

Always separate the connection cables of low voltage accessories from the 115/230 V~ power cables. To prevent any interference whatever, use separate sheaths.



Maximum length of power cables between control unit and motors must not exceed 10 m, using cables of 2.5mm² diameter.

To secure the various components inside of the airtight enclosure, refer to paragraph 15.

4. CONNECTIONS AND OPERATION

4.1. TERMINAL BOARD CN1

4.1.1. POWER SUPPLY 22V

"VAC-VAC" terminals. The secondary circuit of the 24V ~ 50/60 Hz transformer should be connected to this input. Power supplied by the transformer is signalled by the lighting of the "ALIM" LED located under the terminal board.

4.1.2. BATTERIES

"BAT - BAT" terminals. Connect the power cables of the buffer batteries (optional) to these terminals. The control unit is designed to operate with two buffer batteries, with the minimum characteristics shown on the table in paragraph 2. During normal operation, the unit keeps the batteries charged and these start operating if no power is supplied to the transformer.



Power supplied by batteries only should be considered an emergency situation. The number of possible manoeuvres is linked to the quality of the batteries, the structure of the gate to be moved, the time elapsed since power cut occurred, etc, etc.



Observe the battery power supply polarity.

4.1.3. ACCESSORIES

"24V - 24V" terminals. The accessories power cables should be connected to these terminals.



The maximum load of the accessories must not exceed 500 mA.



The output of these terminals is DC - observe the power supply polarity of the accessories.

4.1.4. EARTH

"⊕" terminal. The control unit earthing cable should be connected to this terminal.



This connection is absolutely necessary to ensure a correctly operating control unit.

4.2. TERMINAL BOARD CN2

4.2.1. GEARMOTOR 1

"APM1 - CHM1" terminals. For double leaf applications, connect to these terminals the gearmotor fitted on the leaf which must move first. For single leaf applications, the gearmotor must be connected to these terminals. The maximum load of the gearmotor must not exceed 70W.

4.2.2. GEARMOTOR 2

"APM2 - CHM2" terminals. For double leaf applications, connect to these terminals the gearmotor fitted on the leaf which must move last. For single-leaf applications, nothing should be connected to these terminals. The maximum load of the gearmotor must not exceed 70W.

4.2.3. ELECTRIC LOCK

"ELS - ELS" terminals. The electric lock, if any, with 12 Vdc/~ power supply, should be connected to these terminals. To facilitate release of the electric lock, the over-pushing stroke can be input by enabling parameter "F" (see paragraph 9).



In double-leaf applications, install the electric lock on the leaf where gearmotor 1 is installed.

4.2.4. FLASHING LAMP / COURTESY LIGHT

"LAMP - LAMP" terminals. Both a flashing lamp and a courtesy light can be connected to these terminals, both with 24 Vdc power supply and maximum 15W. To make this output operational, select parameter "G", see paragraph 9.

Flashing lamp operation:

During normal operation, the flashing lamp provides a fixed pre-flashing of 1.5 seconds during both opening and closing. When the gate is open, and the closing safety devices are tripped, the lamp flashes to indicate that a manoeuvre is taking place in the gate movement area. We advise you to connect the flashing lamp before programming, because it indicates its phases. Use a fixed light flashing lamp; flashing is controlled by the control unit.

Courtesy light operation:

The courtesy light stays lighted for a fixed time of 90 seconds from the OPEN pulse, after which it goes OFF. Use a lamp with 24 V power supply and maximum 15W.

4.3. TERMINAL BOARD CN3

4.3.1. MOTOR 1 CLOSING LIMIT-SWITCH

"COMF - FCC1" terminals. Normally closed contact. This is tripped and stops the closing motion of motor 1. The status of this input is signalled by LED FCC1.

4.3.2. MOTOR 1 OPENING LIMIT-SWITCH

"COMF - FCA1" terminals. Normally closed contact. This is tripped and stops the opening motion of motor 1. The status of this input is signalled by LED FCA1.

4.3.3. MOTOR 2 CLOSING LIMIT-SWITCH

"COMF - FCC2" terminals. Normally closed contact. This is tripped and stops the closing motion of motor 2. The status of this input is signalled by LED FCC2.

4.3.4. MOTOR 2 OPENING LIMIT-SWITCH

"COMF - FCA2" terminals. Normally closed contact. This is tripped and stops the opening motion of motor 2. The status of this input is signalled by LED FCA2.



• If no limit-switch is used, jumper connect the inputs.

• The limit-switches cannot be used as the start of the decelerated section.

4.3.5. MOTOR 1 ENCODER

"ENC1" terminal. The signal received from the encoder installed on gearmotor 1 must be connected to this terminal. For operation and activation of the encoder, see paragraph 6.



If no encoder is used, do not, on any account, jumper-connect the inputs

4.3.6. MOTOR 2 ENCODER

"ENC2" terminal. The signal received from the encoder installed on gearmotor 2 must be connected to this terminal. For operation and activation of the encoder, see paragraph 6.



If no encoder is used, do not, on any account, jumper-connect the inputs

In two-motor applications, the encoder must be installed on both motors.

4.4. TERMINAL BOARD CN4

4.4.1. TOTAL OPENING

"COM - OPEN A" terminals. Normally open contact. Connect, to these terminals, any pulse generator (e.g. push-button, key selector, etc.) which, by closing a contact, generates a gate total opening or closing pulse. The operation of this generator is defined by parameter "D", see paragraph 9.



• A total opening pulse always has priority over pedestrian opening.

• To connect several pulse generators, connect the devices in parallel.

4.4.2. PEDESTRIAN OPENING

"COM - OPEN B" terminals. Normally open contact. Connect, to these terminals, any pulse generator (e.g. push-button, key selector, etc.) which, by closing a contact, generates a gate partial opening or closing pulse. In double leaf applications, pedestrian opening corresponds to total opening of leaf 1. In single leaf applications, pedestrian opening corresponds to about 30% of memory-stored total opening.



• A total opening pulse always has priority over pedestrian opening.

• To connect several pulse generators, connect the devices in parallel.

4.4.3. Stop

"COM - STOP" terminals. Normally closed contact. Connect, to these terminals, any safety device (e.g. pressure switch, safety edge, etc.) which, by opening a contact, immediately stops the gate and disables all automatic functions. The status of this input is signalled by the "STOP" LED. The gate resumes its memory-stored cycle only by means of another total or partial opening pulse.



• If no STOP devices are connected, jumper connect the input.

• To connect several STOP commands, connect the devices in series.

4.4.4. CLOSING SAFETY DEVICES

"COM - FSW CL" terminals. Normally closed contact. Connect, to these terminals, any safety device (e.g. photocell, safety edge, pressure switch etc.) which, by opening a contact, affects the gate's closing motion, reversing it to the mechanical stop, or to the opening limit-switch. The status of this input is signalled by LED "FSW-CL".

4.4.5. OPENING SAFETY DEVICES

COM - FSW OP terminals. Normally closed contact. Connect, to these terminals, any safety device (e.g. photocell, safety edge, pressure switch etc..) which, by opening a contact, affects the gate's opening motion, causing it to stop immediately. When the safety device has been reset, the gate resumes its memory-stored cycle. The status of this input is signalled by LED *FSW-OP*.



- If no safety devices are connected, jumper connect the inputs.
- To connect several safety devices, connect the devices in series.

5. INSTALLING THE RADIO CONTROL RECEIVER BOARD

The control unit is designed to house a 5-pin radio-receiver module. Installation procedure: turn off power and fit the module in connector **CN5** on the control unit.



To avoid damaging the receiver and thus irreparably compromising its operation, the receiver must be installed while observing the fitting direction specified in paragraph 13 (connection lay-out).

This done, observe the radio-receiver instructions to store the radio control in the memory.

6. OPERATION WITH ENCODER OR AMPEROMETRIC OPERATION

The control unit has 4 **DIP SWITCHES**, which enable selection of either amperometric operation or operation with Encoder.

Operation with encoder provides greater safety in detecting obstacles and greater repeatability of the deceleration point.



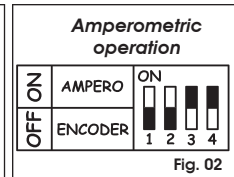
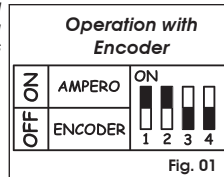
Operation with encoder requires mechanical stops, or limit-switches, for both opening and closing.

To select operation with encoder, position **DIP-SWITCHES 1 and 2 to ON** and **DIP-SWITCHES 3 and 4 to OFF** (Fig.01).

To select amperometric operation, position **DIP-SWITCHES 1 and 2 to OFF** and **DIP-SWITCHES 3 and 4 to ON** (Fig.02).



For a correct programming procedure of the control unit, carry out this operation before programming the control unit because it radically modifies its operation.



7. CONTROL LEDS

LEDS	ON	OFF
ALIM	Power supply by toroidal transformer	Power supplied by batteries or no power supplied
FCC1	Motor 1 closing limit-switch not tripped	Motor 1 closing limit-switch tripped
FCA1	Motor 1 opening limit-switch not tripped	Motor 1 opening limit-switch tripped
FCC2	Motor 2 closing limit-switch not tripped	Motor 2 closing limit-switch tripped
FCA2	Motor 1 opening limit-switch not tripped	Motor 2 opening limit-switch tripped
STOP	Stop command not activated	Stop command activated
FSW-CL	Closing safety device not tripped	Closing safety device tripped
FSW-OP	Opening safety device not tripped	Opening safety device tripped



Indicated in bold: status of LEDs with gate closed, control unit powered, and both limit-switches installed.

- If the limit-switches are not used, the relevant contacts must be jumper connected and LEDs **FCC1 - FCA1 - FCC2 - FCA2** must be lighted.
- If no **STOP** devices are connected, jumper connect the input. The **STOP** LED must be lighted.

8. OPERATION OF DISPLAY

The control unit has a handy display for viewing and programming the operating parameters. Furthermore, it constantly shows gate status during normal operation.

When operating parameters are being displayed and programmed, the display shows the selected parameter on the left, and its set value on the right. Fig. 03 shows the viewing example of parameter "A" at value "2".



During normal operation, the display shows gate status. The displayed values are indicated on the following table:

DISPLAY VALUE	GATE STATUS
- -	Gate at rest
□ P	Gate opening
⏸ c	Gate open in pause status (Only with automatic re-closure enabled - see next paragraph)
□ L	Gate closing

9. ADJUSTING THE OPERATING PARAMETERS



Before you begin adjusting the operating parameters, select the type of operation for the control unit: with or without encoder (see paragraph 6).

To access operating parameter adjustment, follow the instructions below:

- 1- When you have made all the necessary connections, power up the system and check if all the signalling LEDs are in the situation specified in paragraph 7.
- 2- The display shows value "---".
- 3- Press and hold down push-button **P2** until the display shows the name and value of the first parameter.
- 4- Press push-button **P1** to change the value of the parameter.
- 5- To move on to the next parameter, press push-button **P2**.
- 6- When 60 seconds have elapsed without any key being touched, the control unit exits the adjustment mode. You can manually exit the adjustment mode by scrolling all the parameters with push-button **P2**. When the display shows value "---", you have returned to normal operation.

The following table summarises all settable parameters and the assignable values.

DISPLAY	DESCRIPTION
Adjusting the sensitivity of the electronic clutch	
A1	Minimum motor force
A2	Medium-low motor force
A3	Medium-high motor force
A4	High motor force
Leaf 2 delay: this parameter enables you to select the offset time of the two leaves.	
b1	1.5 seconds offset
b2	3 seconds offset
b3	6 seconds offset
b4	10 seconds offset
Automatic Reclosure: this function enables or disables automatic gate reclosure.	
c0	Disabled
c1	Enabled
Operation of OPEN A command: this function determines the behaviour of the OPEN A (total opening)push-button.	
d0	Opens / Closes / Opens.....
d1	Opens / Stops/ Closes / Stops / Opens.....
Condo function: if this function is enabled while the gate is being opened, the start command is inhibited.	
E0	Disabled
E1	Enabled
Over pushing stroke: if you enable this function, at every OPEN pulse, the leaf on which the electric lock is installed starts its closing movement for a few seconds. This facilitates release of the electric lock.	
F0	Disabled
F1	Enabled
Courtesy light/ Flashing lamp: with this parameter, you can select the type of output from the LAMP - LAMP terminals, selecting from among flashing lamp and courtesy light.	
G0	Flashing lamp
G1	Courtesy light (active for 90 seconds)

DISPLAY	DESCRIPTION
Deceleration point percentage: this parameter is used to set the length of the decelerated section, selecting from the two set values.	
H0	40% of maximum memory-stored opening
H1	20% of maximum memory-stored opening
Speed during deceleration phase: this parameter is used to set motor speed during the deceleration phase, selecting it from the two values.	
i0	High
i1	Low
Limit-switch operation: this parameter enables you to select operation with or without the limit-switch.	
L0	Operation without limit-switch
L1	Operation with limit-switch
Number of motors: this parameter is used to select the type of gate: with one leaf or with two leaves	
n1	Single-leaf gate, only one motor connected
n2	Double-leaf gate, two motors connected

10. PROGRAMMING

 Before you begin programming, select the type of operation for the control unit: with or without encoder (see paragraph 6).

During the programming procedure, the control unit memory-stores the opening, closing mechanical stop points and any pause time before re-closure.

- 1- Release the gearmotors, locate the leaves at half open point, and re-lock the operators.
- 2- Power up the control unit and check if value "--" is shown on the display.
- 3- Press and hold down push-button **P2** until the display shows the first parameter and relevant value.
- 4- Give an **OPEN A** command with any device connected to this input: the display shows value "P-", and the leaves begin to move. The first manoeuvre performed by the leaves must be closing. If this does not happen, stop the gate movement with a reset pulse. To do this, make a bridge between the two **"RESET"** PINS, using the connector (see Fig.04). Cut power and put the connector back in its normal position (Fig. 04). Then change over the wires of the motor/s, which had performed the opening operation. Resume the programming stage from point 1.
- 5- When the closing mechanical stop point is reached, the gearmotors pause for about 2 seconds, and then restart with a total opening manoeuvre up to the opening mechanical stop point or up to the relevant limit-switch.
- 6- If automatic reclosure was not enabled, this means programming has finished, otherwise the control unit begins counting pause time.
- 7- When the required time has elapsed, give another **OPEN A** pulse, and the gate will begin to close.
- 8- When the closing stop has been reached, programming has terminated, and the display shows value "--".

-  • The display shows value "--" during the entire programming procedure.
- The flashing lamp stays lighted on a fixed light during the entire programming time.
- Leaf movement is decelerated during the programming procedure.

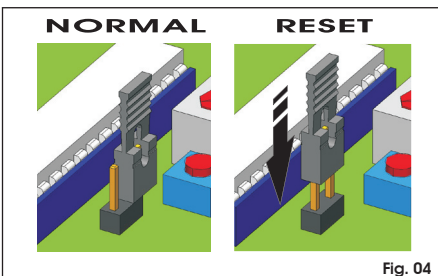


Fig. 04

11. OPERATION OF ELECTRONIC CLUTCH

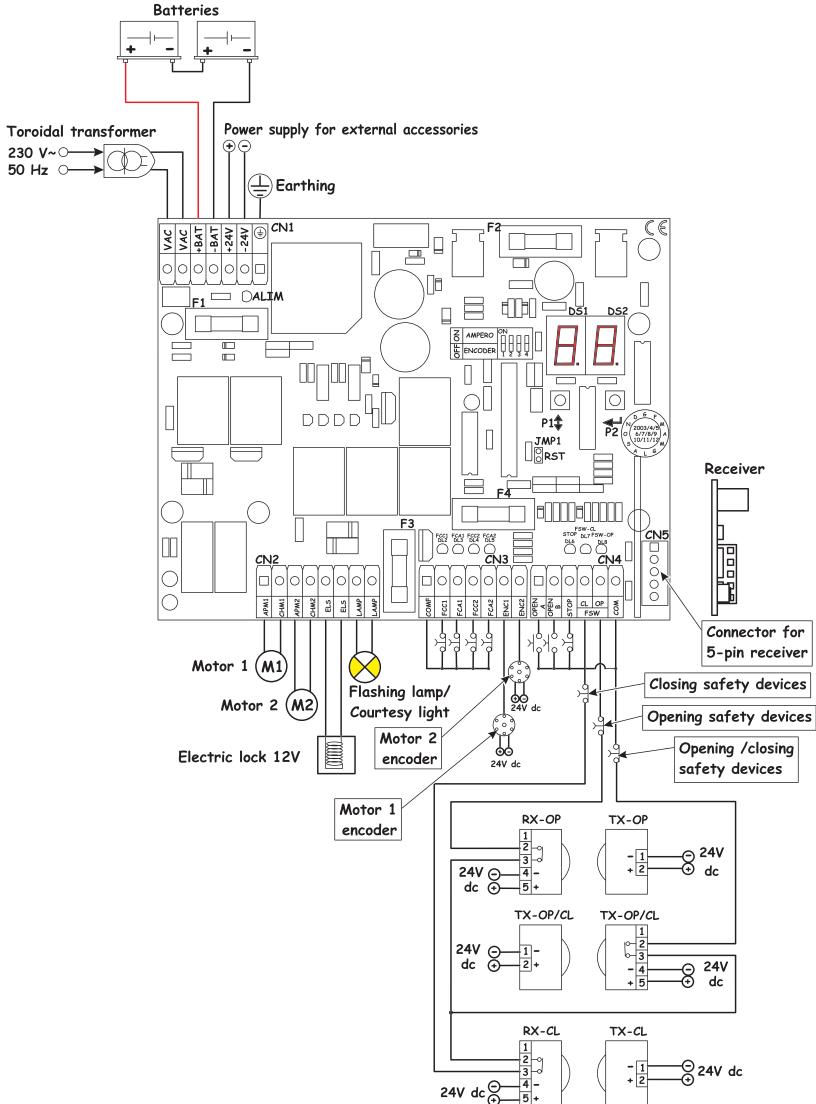
This is a very important device for reasons of safety. Its setting stays unchanged long-term, without wear. It is active during both closing and opening. When it operates, it reverses gate movement without disabling automatic closing if activated.

If the clutch operates several consecutive times during the closing movement, the control unit goes into **STOP** status, disabling any automatic command. If the clutch operates several consecutive times, this means that the obstacle remains and it could be dangerous to perform any manoeuvre. To restore normal operation, the user must give an **OPEN A** / **OPEN B** pulse.

12. PROTECTION FUSES

FUSE	PROTECTION	FUSE	PROTECTION	FUSE	PROTECTION	FUSE	PROTECTION
F1=T10A 250V - 5x20	Power supply 24V~	F2=T0.63A 250V - 5x20	Supply to accessories and battery- charger	F3=R0.63A 250V - 5x20	Flashing lamp output	F4=R3.15A 250V - 5x20	Electric lock output

13. CONNECTION LAY-OUT



14. FUNCTION LOGICS

Automatic "A" logic C=1 d=0 E=0						
Pulses						
Gate status	Open A	Open B	Stop	Opening safety devices	Closing safety devices	OP/CL safety device
Closed	Opens the leaf and re-closes after pause time	Executes leaf partial opening and re-closes after pause time	No effect (OPEN disabled)	Disables OPEN commands	No effect	Disables OPEN commands
Open on pause	Reloads pause time	Closes the gate immediately	Stops operation	No effect	Stops closure. On release, closes after 5 sec., if pause time has elapsed	Locks pause time, and when released, closes after 5 sec.
Closing	Reverses gate motion	No effect	Stops operation	No effect	Reverses motion	Stops operation and reverses on release
Opening	Reverses gate motion	No effect	Stops operation	Stops operation and resumes on release	No effect	Stops operation and resumes on release
Stepped Automatic "AP" logic C=1 d=1 E=0						
Pulses						
Gate status	Open A	Open B	Stop	Opening safety devices	Closing safety devices	OP/CL safety device
Closed	Opens the leaf and re-closes after pause time	Executes leaf partial opening and re-closes after pause time	No effect (OPEN disabled)	Disables OPEN commands	No effect	Disables OPEN commands
Open on pause	Reloads pause time	Closes the gate immediately	Stops operation	No effect	Stops closure. On release, closes after 5 sec., if pause time has elapsed	Locks pause time, and when released, closes after 5 sec.
Closing	Stops gate motion and opens on next pulse	No effect	Stops operation	No effect	Reverses motion	Stops operation and reverses on release
Opening	Stops gate motion and closes on next pulse	No effect	Stops operation	Stops operation and resumes on release	No effect	Stops operation and resumes on release

Semi-automatic "E" logic C=0 d=0 E=0						
Pulses						
Gate status	Open A	Open B	Stop	Opening safety devices	Closing safety devices	OP/CL safety device
Closed	Opens the leaf	Executes partial opening	No effect (OPEN disabled)	Disables OPEN com- mands	No effect	Disables OPEN com- mands
Open	Closes	Closes the gate	No effect (OPEN disabled)	No effect	Disables OPEN com- mand and, on release, re-closes after 5 sec..	Disables OPEN com- mand and, on release, re-closes after 5 sec..
Closing	Reverses gate motion	No effect	Stops operation	No effect	Reverses gate motion	Stops operation and reverses on release
Opening	Reverses gate motion	No effect	Stops operation	Stops operation and resumes on release	No effect	Stops operation and resumes on release
Stepped semi-automatic "EP" logic C=0 d=1 E=0						
Pulses						
Gate status	Open A	Open B	Stop	Opening safety devices	Closing safety devices	OP/CL safety device
Closed	Opens the leaf	Executes partial opening	No effect (OPEN disabled)	Disables OPEN com- mands	No effect	Disables OPEN com- mands
Open	Closes	Closes the gate	No effect (OPEN disabled)	No effect	Disables OPEN com- mand and, on release, re-closes after 5 sec..	Disables OPEN com- mand and, on release, re-closes after 5 sec..
Closing	Stops gate operation and opens on next pulse	No effect	Stops operation	No effect	Reverses gate motion	Stops operation and reverses on release
Opening	Stops gate operation and opens on next pulse	No effect	Stops operation	Stops operation and resumes on release	No effect	Stops operation and resumes on release
Condo "D" logic C=1 d=0 E=1						
Pulses						
Gate status	Open A	Open B	Stop	Opening safety devices	Closing safety devices	OP/CL safety device
Closed	Opens the leaf and re- closes after pause time	Executes leaf partial opening and re-closes after pause time	No effect (OPEN disabled)	Disables OPEN com- mands	No effect	Disables OPEN com- mands
Open on pause	Reloads pause time	Closes the gate immediately	Stops operation	No effect	Disables OPEN com- mand and, on release, re-closes after 5 sec..	Disables OPEN com- mand and, on release, re-closes after 5 sec..
Closing	Reverses gate motion	No effect	Stops operation	No effect	Reverses gate motion	Stops operation and reverses on release
Opening	No effect	No effect	Stops operation	Stops operation and resumes on release	No effect	Stops operation and resumes on release

15. HOW TO SECURE THE BOARD

The outdoor enclosure is designed to house the control unit, the toroidal transformer and any buffer batteries (Optional).

To secure the toroidal transformer and the board support, consult the specific instructions.

To secure the control board, follow the instructions below:

1. Position the supplied spacers (Fig.5 ref.①) on the columns identified by the following letters: **A-C-I-P-Q-G**.

2. Secure the board using the supplied screws (Fig.5 ref. ②).



*The spacers positioned next to letters **A & P** serve only for supporting the board.*

3. Wire up according to the connection lay-out (paragraph 13).

4. To position and wire the battery kit, refer to the relevant instructions.



If you are using the battery kit, you MUST clear the pre-perforated hole in the lower part of the enclosure (Fig.6 ref. ①) as requested by current safety laws.

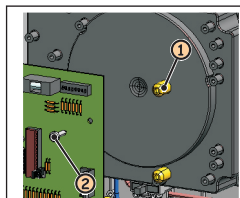


Fig. 05

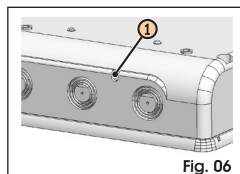


Fig. 06

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